

## The Unicorn's Horn



*The Sense of Sight*  
from [The Lady and the Unicorn](#)  
tapestry collection

A passage out of *Fisherman's Guide to the Cosmic Order* that has always stood out to me ever since I first read the book is the following section on page 41 in Chapter 4 - Clouds:

...This time a tunnel-like hole opened in front of me down into the void, like looking into another bottomless well that had access into and through the energies of the void. There was again a powerful polarization throughout my body as I saw a very rapid zigzag streak of light projected down inside the tunnel to inscribe what appeared to be an irregular six-pointed figure that vanished as quickly as it was formed.

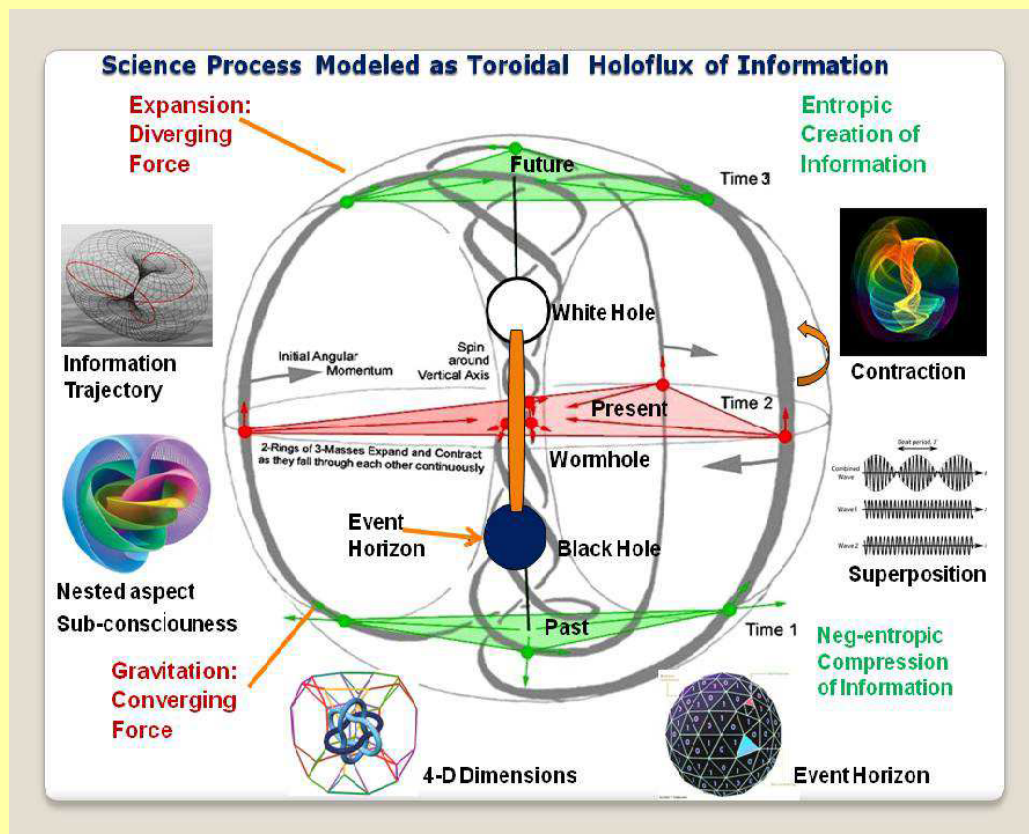
Of course, the “six-pointed figure” seems to be the hexad of The Enneagram, or to relate it to Projective Geometry, Pascal's “Hexagrammum Mysticum”.

Moreover, what intrigued me most was the fact that a Center is connected to The Universal Periphery by means of a vortex that is accessed through a particular geometric pattern. This has also been the case within some of my own spiritual experiences.

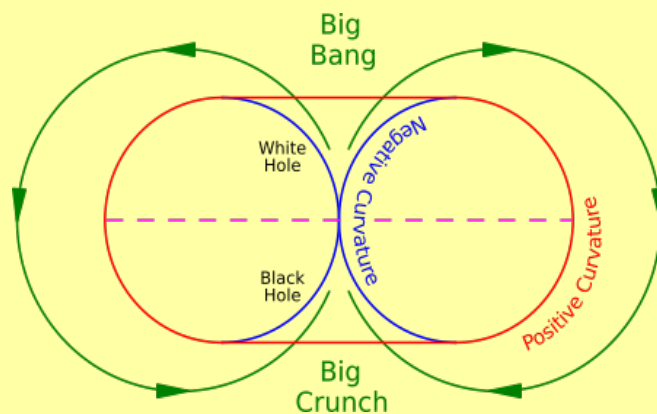
I wanted to take a moment to meander through the forest of thought together to look for a few of the things that I find special. Hopefully it doesn't seem rambling or strange. You are probably already familiar with a lot of this already. Like most of the things that I send, there are many links. A small number of them are repeats from past letters. Some of the new ones are there for quick reference, while others are for later perusal at your leisure. I think you'll enjoy some of them.

To begin...

I was thrilled with how you related Dirac's continuous creation cosmology to a black hole-white hole pair and to T-symmetry:



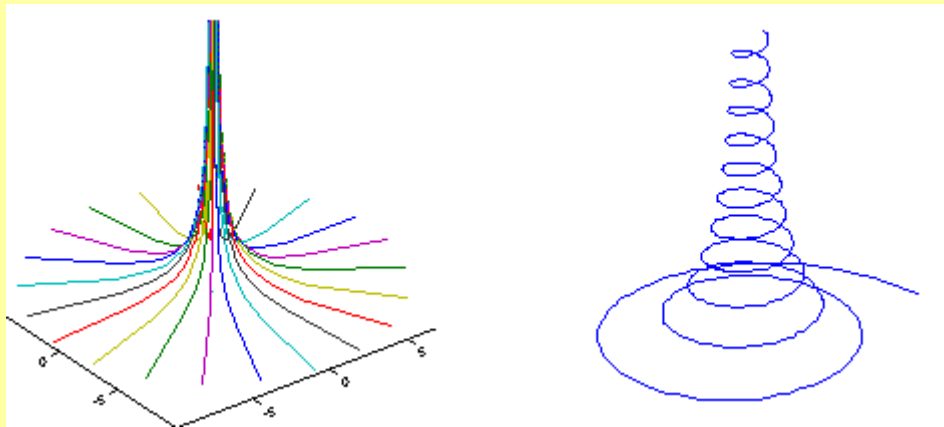
I look at it in exactly the same way:



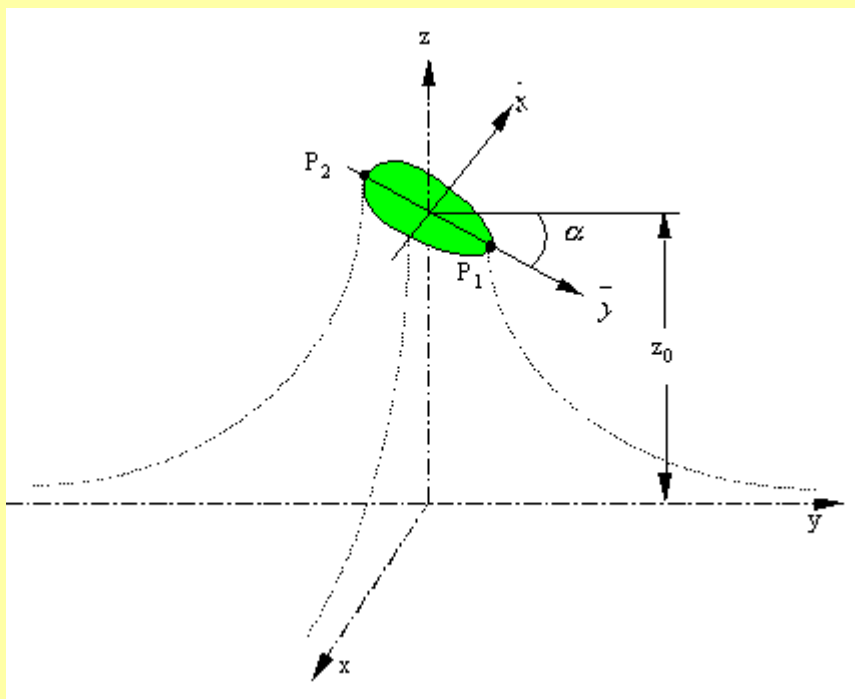
Of course, we both know that the spiral vortex is “the horn of the unicorn”. ☺

In [Part 4: The Unified Field](#) of *It's Vortices! Vortices All The Way Down!*, I cover one way of deriving the shape of the horn. It seems Euclidean in that I rotate a rectangular hyperbola around an axis within a regular Cartesian coordinate system, but a kind of Projective Geometry is implied. I assume that the two ends of the hyperbola, the asymptotes, eventually meet at infinity to form a horn torus, like the one depicted in cross-section within the above image. I also noted a musical pattern within the rectangular hyperbola based on the work of Viktor

Schauberger. Quite some time ago, I had come across [an article by Christian Lange](#) that highlights this same relationship:



Kepler's Laws of Planetary Motion, which describe how the velocities of planets change throughout their orbit around the Sun, are often approximated with an ellipse. One way of deriving an ellipse is through a conic section. But what do we get when we take a section of a *hyperbolic* cone like the one above?



We end up with an egg shape. To quote [with some editing]:

...Kepler used the elliptic equations only for simplifying the calculations, obtaining an approximation, and telling everybody that, in reality, planets are following egg-curves. Today, most scientists think that the planets move on elliptic curves, even if it is well known that the gravitational potential of the sun is hyperbolic.

An egg has one focus, unlike an ellipse with two foci. If the sun is one of these foci, then what is the other one? It is [a solar system of nested golden eggs](#).

Christian Lange goes on to mention some other interesting correspondences within the rest of the article, but I simply want to highlight this relationship between vortices, eggs, and Non-Euclidean Geometry.

As Non-Euclidean Geometry developed throughout the 1800s and early 1900s, several spiritual interpretations started to form around the idea of “higher dimensions”. A famous example is the work of [Charles Hinton](#). Perhaps less well-known is the work of Rudolf Steiner, a German mystic who founded [The Anthroposophical Society](#).

In [his lectures on "The Fourth Dimension"](#), he claims that mathematical reasoning can help one to experience more subtle aspects of reality, like the so-called “Astral world”. He even mentions [the importance of learning to perceive of things in reverse](#) because time seems to move backwards within these other “planes of existence”!

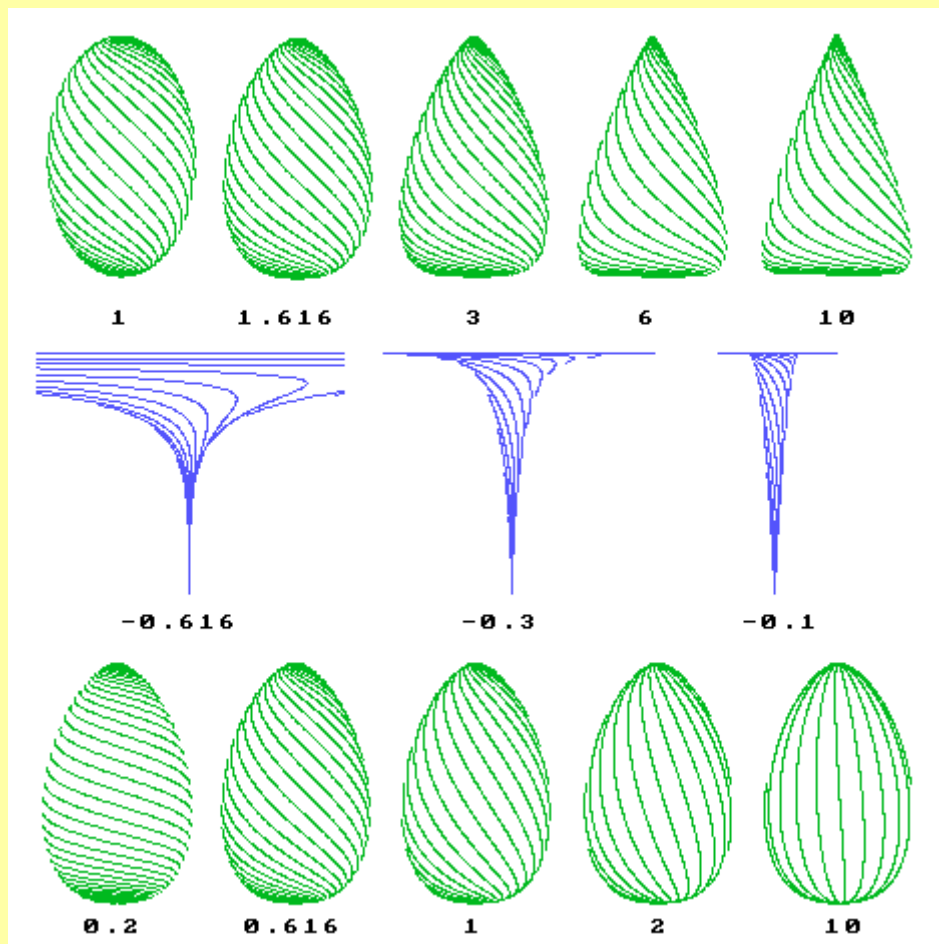
It is interesting to note the relation of the term “astral” to “star” (and thus, to “heaven”). Likewise, many mystics speak of an “[astral body](#)” that can be sensed as a layer within the “aura” around someone’s physical body. Impressions are said to appear within the aura before they become manifest within or through the person. There’s that time-reversal again! In this case, we are looking at it within the spiritual domain rather than the scientific (like the examples that I shared before on the measurement of L-fields with vacuum tube voltmeters, pre-stimuli responses within the nervous system, a “law of syntropy” derived from the negative solutions of the energy-momentum equation, etc.)...Science and spirituality are often two complementary halves of One whole...

Of particular importance to Steiner in honing this subtle sense was the study of Projective Geometry. Many mathematicians built upon these concepts (such as George Adams, [Olive Whicher](#), Louis Locher-Ernst, Lawrence Edwards, [Nick Thomas](#), et al.). They also tried to apply it to biology and to [physics](#). Sadly, much of their work remains obscure.

There used to be [a wonderful website](#) that provided [a summary](#) of some aspects of this form of mathematics, but it seems to be long gone. Someone has made [a PDF backup of it](#) though.

One prevailing theme is the idea of geometry taking place within two distinct “spaces” simultaneously. One is the traditional Euclidean Space that seems to expand infinitely outward towards a Universal Periphery. The other is a Projective “Counterspace” that seems to contract infinitely inward towards a Universal Center.

George Adams tried to connect these two spaces with a particular construction referred to as a “[Path Curve](#)”. Then, Lawrence Edwards used these to model the curvature of organic forms, particularly the buds of various plants and vortices within air and water. [As an alchemical aside, he also noted a very interesting correlation between plant growth and planetary conjunctions.] Clopper Almon gives [an algebraic treatment of this work](#), and Douglas Baker [uses it to describe bird’s eggs](#). Again, we encounter the vortex and egg...



Lawrence Edwards formulated an equation with two parameters symbolized by the Greek letters lambda and epsilon. When lambda is positive, it generates an egg shape that becomes progressively more conical as lambda increases (like in the top row of the above image). When lambda is negative, it generates a vortex (like in the middle row). Meanwhile, the value of epsilon controls the degree of the spiral along the surface of either shape (like in the bottom row).

[Eric Dollard](#) has fruitfully applied the concepts of Counterspace and Reverse Time to electrical engineering. In general, these become very important when trying to understand [how electricity appears throughout Nature](#). Most of modern physics has a tendency to approach these same phenomena through somewhat abstruse mathematical models instead of direct experimentation and physical analogies, but they are still there. Meanwhile, Projective Geometry is nearly synonymous with the history of architectural painting and [the Subjective experience of our own visual system](#). Some have gone a step further and suggested a fundamental relationship to consciousness itself, as in [The Projective Consciousness Model](#).

So, if individual humans are part of a larger holon, does the Earth also have an “aura”? Bob sometimes mentions The Planet Mind. Similar to how the L-field is akin to the Host of the human being, an electrical pattern that serves as the physical manifestation of the “organization of an organism”, the fields that envelope the Earth act in the same way on a larger scale. It has been referred to as the “Noosphere”. One of the best summaries that I’ve come across is from the introduction to José Argüelles’ book [Manifesto for the Noosphere - The Next Stage in the Evolution of Human Consciousness](#). To quote [with some extra links and highlighting added by me]:

We must enlarge our approach to encompass the formation taking place before our eyes ... of a particular biological entity such as has never existed on earth - the growth, outside and above the biosphere, of an added planetary layer, an envelope of thinking substance, to which, for the sake of convenience and symmetry, I have given the name of the Noosphere. – Pierre Teilhard de Chardin, *The Future of Man*

*Manifesto for the Noosphere* is the result of forty years of study, contemplation, investigation, and synthesis. While the noosphere may be beyond the grasp of conventional science, it is a deep and pervasive intuition that has gripped the minds of scientists, philosophers, poets, and artists since the concept first emerged in 1926. It is an evolutionary concept posited by studies in both biogeochemistry and paleontology. It is a whole-systems paradigm that melds prophecy and analysis of current world trends. It is a perception that the transformation of the biosphere is inevitably leading to a new geological epoch and evolutionary cycle, and it is due to the impact of human thought on the environment that this new era - the Noosphere - is dawning.

The term “noosphere,” referring to the mental sheath or envelope of thought that encompasses the Earth, is derived from *nous*, the Greek word for “mind.” The presentiment of the noosphere fully awakened in me in 1969 upon seeing the whole Earth from space on the television. Soon after, I organized the First Whole Earth Festival, and together with my students at the University of California, Davis, we transformed the central quadrangle of the campus into the “global village” - all in anticipation of the first Earth Day set to occur a month later, on April 22, 1970.

Around the same time I had a fruitful correspondence with R. Buckminster Fuller, who first suggested to me the presence of an information storage and retrieval system existing as some kind of psychic field or thought belt around the planet. It was by this means, Bucky wrote me, that he could converse with the pre-Socratic philosophers as he strolled down the beach.

The First Whole Earth Festival roused the interest of planetary philosopher and astrologer Dane Rudhyar, whose book [\*The Planetarization of Consciousness\*](#) (1970) had just been published. A warm friendship immediately ensued, and among other things, Rudhyar exposed me to the work of University of Pittsburgh physicist Oliver Reiser. In his monumental volume, [\*Cosmic Humanism\*](#) (1966), Reiser proposes a vision of the noosphere as consisting of a psi field between and in resonance with the two radiation belts defining Earth's electromagnetic field. The psi field, a DNA-informed “World Sensorium,” Reiser further hypothesizes, consists of two halves - an Eastern (intuitive) and a Western (analytical) - which function

holonomically like the two hemispheres of the human brain.

I amplified Reiser's notion in my book [\*The Transformative Vision\*](#) (1975), where I link the two principles governing the history of human consciousness - psyche and techne - to the two sides of the human and planetary brain. While acknowledging the vision of Teilhard de Chardin, among many others, this book was also influenced in its perspective by Hopi, Hindu, Mexican, and Mayan prophetic traditions. But it is with the notion of the Earth and its hemispheres - Eastern and Western as well as Northern and Southern - that *The Transformative Vision* builds on the idea of a higher unifying principle of planetary consciousness governing the progression of history and the world ages in general.

In [\*Earth Ascending\*](#) (1984), elaborating greatly on Teilhard de Chardin's notion of "planets with noosphere," I put forward the existence of a psi bank - the noosphere's coded information storage and retrieval system and DNA timing program, located between and in resonance with (as Reiser had proposed) the two radiation belts of Earth's electromagnetic field. The psi bank has been key to my later work, including *The Mayan Factor* (1987), *The Call of Pacal Votan: Time Is the Fourth Dimension* (1996), and *Time and the Technosphere* (2002).

My studies of [the mathematics of the Mayan calendar](#) and subsequent discovery of the Law of Time as the universal principle of synchronization, as well as exposure to [the work of Vladimir I. Vernadsky](#), further deepened my comprehension of the noosphere. Through a number of visits to Russia and my dialogue with leading-edge Russian scientists, I saw how widespread the concept of the noosphere is in Russian intellectual life. By 2003 I had become an acting member of the Moscow-based Noosphere Spiritual Ecological World Assembly. Having already convened the First and Second Planetary Congresses of Biospheric Rights in Brasilia (1996, 2006), I proposed the First Noosphere World Congress to occur in Bali, 2009. This event evolved into a number of synchronized bioregional Noosphere Congresses convened on July 22, 2009. From this experience came my website.

All this would not have been if a Jesuit priest, Pierre Teilhard de Chardin, had not coined the concept and word in a burst of inspiration while serving as a stretcher-bearer in the front-line trenches of the First World War. As a noted paleontologist, Teilhard de Chardin went on to write extensively of the noosphere in books like [\*The Future of Man\*](#) (1959) and [\*Man's Place in Nature\*](#) (1956). In 1926 he and his Jesuit colleague and fellow philosopher Edouard Le Roy met with a Soviet geochemist, Vladimir I. Vernadsky, to jointly agree on the meaning of the

concept *noosphere*. This meaning is [spelled out very simply and clearly by Vernadsky](#), who wrote at the end of World War II:

*The historic process is changing dramatically before our eyes ... Mankind taken as a whole is becoming a powerful geological force. Humanity's mind and work face the problem of reconstructing the biosphere in the interests of freely thinking mankind as a single entity. This new state of the world we are approaching without noticing it is the "Noosphere."*

Just as the biosphere is the unified field of life and its support systems - the region for the transformation of cosmic energy on Earth, to use Vernadsky's phrase - so the noosphere is the unified field of the mind, the psychic reflection of the biosphere. Because we as a species, the aggregate of consciousness-bearing cells of the evolving Earth, are not yet awake to our role as a planetary organism, so too the noosphere is not yet fully conscious. [When humanity becomes conscious of itself as a single organism](#) and unites to activate the noosphere, [we will find the collective resolve and will to reconstruct the biosphere and divert the energy of the human race from a path of destruction based on a mechanized abstraction from nature](#) to a new harmonic order of super-organic reality based on an entirely different state of consciousness than has yet existed on Earth...

Several others have posited the existence of similar ideas under different names (e.g.: the biologist Lyall Watson's notion of a "[Lifetide](#)", based on Carl Jung's "Collective Unconscious"). I've been wanting to write [an article on The Planet Mind](#) that goes into these topics with more depth, but it only contains a brief note that relates it to System 3 at the moment.

There is [a lot](#) that can be said about all of this, but I will focus in on only one aspect for now: Bob has repeatedly demonstrated the concept of Biospheric Resonance, giving examples of many seemingly different animals evolving in parallel, converging and diverging in expression throughout the Biosphere. José Argüelles attempted to do a similar study with human history. In *Earth Ascending*, he makes reference to what he calls "Seven Pristine Streams", ancient cultures that were geographically isolated from one another, yet share certain fundamental characteristics, like reoccurring "[cultural universals](#)" appearing at the beginning of every civilization. This is probably not too surprising as [the expression of human culture has roots in our own biological development](#).

In contradistinction to Bob's logical and scientific approach in *Downsizing Darwin* though, a large portion of *Earth Ascending* is made up of artistic diagrams with intuitive descriptions revolving around the topics of geomancy, magic squares, the I Ching, the Mayan calendar, and so on. It is fertile ground for contemplation.

However, [a lot of research](#) has also been compiled that points to measurable aspects of the "Noosphere". There are various mechanisms by which long-distance biological communication could possibly occur (such as [Biophotons](#), [Fröhlich](#)

[frequencies](#), etc.). If one is willing to delve into the “fringe” and the “alternative”, there are many others. I will offer one hypothesis here. First, we will approach it through the symbolic...

To quote the now defunct website [sacred-geometry.com](#) [with some slight editing and extra links added by me]:



The Crooked Soley Crop Circle appeared at night on August 27, 2002. It was the very last major formation of the [2002 season](#).

[...]

The Crooked Soley formation measured about 300 feet across, and the pattern was made up of 1296 ‘curved squares’. In 504 of these squares the crop was left standing, and they depicted the double helix DNA spiral. The other 792 curved squares were flattened crop that enabled the DNA pattern to stand out in relief.

These are the numbers of sacred geometry. If you multiply the number of standing crop squares 504 by 10, you get [5040](#), the [number used by Plato](#) for the planning of his ideal city. The recommended number of citizens for the ideal city is 5040, and all the institutions in the city are based on that number, as is the physical radius of the city as well. Plato took this number to be the key for a divinely ordered Creation.

In the fascinating book, *Crooked Soley: A Crop Circle Revelation*, the authors point out that 5040 is also a most significant and

auspicious number in numerology and arithmetic. It is factorial 7 (7!), which is the product of  $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7$ . It is also the result of  $7 \times 8 \times 9 \times 10$ . It therefore occupies a pivotal position in systems based on real numbers that impact every facet of life. The number Seven has unique qualities that make it the natural symbol of the universal soul. Philosophers call it the Virgin and the eternal feminine in nature.

[...]

On the lower plane, where numbers multiply, grow, and elaborate, the archetypal Seven is represented by its factorial, 5040. St. John, as well, in [his vision of the Heavenly City](#), describes its radius as 5040, the same number that Plato adopted as the key to a divinely ordered Creation, the 'heavenly pattern.'

(And by the way, if you multiply the number of curved squares of the flattened crop 792 by 10, you get 7920, which is the result of  $8 \times 9 \times 10 \times 11$ ...

These numbers have repeatedly shown up within my studies. For example, in Vortex-Based Mathematics (VBM), much of it revolves around an archetypal pattern known as "The ABHA Torus". [In more "traditional" mathematics, it is related to [the family of "Clifford Tori"](#).] The word "ABHA" is [a term used within the Baha'i religion](#). We might think of it as synonymous with "The Kingdom of Heaven", whereas its inverse, "BAHA", is the earthly realm. The surface of this torus is often treated as a tessellation made up of 1296 diamond-shaped tiles, just like the number of squares within the crop circle. Similarly, in Syndex Number Dynamics, there is a spiral figure called "Holotome E" that is based on 2520, which is the first number divisible by all of the Base-Ten numerals and half of 5040. The connections that VBM and Syndex make between the different branches of mathematics are akin to byways into [Langland's Program](#).

Why does the crop circle use these specific numerical resonances to depict [a supercoil of DNA](#) though? And further, how do they relate to the organism? I am again reminded of the relationship between triplets and turns described within [Reading the Green Language of Light](#). There is [an underlying geometry to DNA's structure](#) and these numbers are somehow key in its link to the environment.

...Now, let's look at some biophysics. To quote the article [The Soul in Crooked Soley](#) by Bradley York Bartholomew [with an extra link added by me]:

In their book, *Crooked Soley: A Crop Circle Revelation* Allan Brown and John Michell develop an argument that this particular crop circle represents direct communication from a divine intelligence. Not just any superior intelligence, they say it is the divine intelligence that created the Universe. We are not talking here about UFOs or aliens, nor about a superior intelligence that

exists elsewhere in the Universe. We are talking about the Goddess of all Creation. This is a revelation indeed. In the Crooked Soley crop circle that went down in the wheat fields of Crooked Soley towards the end of August 2002, we find the direct affirmation not only that there is a God, a Universal Creator, but we will find when we take their argument to its logical conclusion, that we are also told specifically what this God is and where it is to be found.

[...]

There is now a great deal of research material that indicates a divine or creative intelligence in the DNA. There are the findings of the Russian molecular biologists, Dr. Pjotr P. Garjajev and Dr. Vladimir Popinin, that the DNA has a mysterious resonance that continues to manifest even after the DNA sample has been physically removed. This has been termed the DNA Phantom Effect. These same researchers have found that the DNA reacts to voice activated laser light provided the correct frequency is found. Using these methods it is possible to actually communicate with the DNA, as well as change the information patterns in the DNA. The Russian researchers have concluded that the DNA acts as a solitonic/holographic computer, and that it acts as a superconductor at body temperature. In addition it is known that the DNA is emitting light photons, and it has been concluded that this resonance and these emissions of the DNA are capable of interacting with conventional electromagnetic forces, the natural magnetic resonance of the Earth, as well as the brain waves of living creatures.

In support of these findings, the Finnish physicist, Matti Pitkänen, has constructed a physical theory that treats the external material world as synonymous with the consciousness of the observer. Fluctuating magnetic wormholes in the DNA are capable of creating a seamless connection with the conventional electromagnetic forces in the external world thus presenting a physical theory of life as a unified magnetic force field. Grazyna Fosar and Franz Bludorf in their book *Vernetzte Intelligenz* describe the findings of the Russian molecular biologists, as well as the theories of Matti Pitkänen, and they conclude that at the level of the DNA there is hypercommunication amongst all sentient beings, a hypercommunication that effectively sets up a networked intelligence that underpins all life in the Universe. By hypercommunication is meant instantaneous communication - zero time lag. At the level of the DNA, in this intelligence network, the concepts of time and space are meaningless, and there can be an instantaneous exchange of information throughout the entire universe. The substratum of the physical universe is simply hypercommunicating information contained and transported via fluctuating magnetic wormholes in the DNA.

Needless to say, by virtue of these theories, all life and all events and phenomena in the external world are merely composites of [a global consciousness that is the Universe](#).

Matti Pitkänen does research into [Geometrodynamics](#). His website has [many of his books available for free](#). Grazyna Fosar and Franz Bludorf's book [Vernetzte Intelligenz](#) (or *Networked Intelligence*) is readily available in German, but I haven't read it yet. There are [English translations of some sections](#) floating around on the Internet though, and I've come across similar concepts throughout my research.

For example, VBM suggests that the sugar phosphates that make up the backbone of the DNA generate electromagnetic vortices within the major groove that function like miniature wormholes according to the Kerr-Newman metric. It is interesting to note the relationship of this to the Hopf Fibration through Taub-NUT space. To [quote](#):

When Roy Kerr developed the Kerr metric for spinning black holes in 1963, he ended up with a four-parameter solution, one of which was the mass and another the angular momentum of the central body. One of the two other parameters was the NUT-parameter, which he threw out of his solution because he found it to be nonphysical since it caused the metric to be not asymptotically flat, while other sources interpret it either as a gravomagnetic monopole parameter of the central mass, or a twisting property of the surrounding spacetime.

I imagine that these micro-wormholes can function as points where time-reversed information “inverts” and starts to become physically manifest. I wonder if something similar happens within the helical channels of microtubules, not to mention within many other places at various levels of scale. ["Superradiance" occurring within networks of tryptophan](#)? I feel as if we are getting very close to the biological basis for how information is drawn out of the Quantum Sensorium.

[ER = EPR](#) is like the theoretical physics equivalent of saying, “Form is Void, and Void is Form.” If there is both a Universal Center and Universal Periphery that are intimately connected, then ultimately, there is no separation. There is just Oneness. System 1.

Even the most basic use of language seems to point to every Particular Form being continually connected to Universal Ideas. Bob states this in reference to our ability to generalize into archetypes, and this was well understood by ancient peoples. To quote pages 21-22 of *The Trivium* by Miriam Joseph:

The intellect through abstraction produces the concept. The imagination is the meeting ground between the senses and the intellect. From the phantasms in the imagination, the intellect abstracts that which is common and necessary to all the phantasms of similar objects (for example, trees or chairs); this is the essence (that which makes a tree a tree or that which makes

a chair a chair). The intellectual apprehension of this essence is the general or universal concept (of a tree or a chair).

A general concept is a universal idea existing only in the mind but having its foundation outside the mind in the essence which exists in the individual and makes it the kind of thing it is. Therefore, a concept is not arbitrary although the word is. Truth has an objective norm in the real.

A general concept is universal because it is the knowledge of the essence present equally in every member of the class, regardless of time, place, or individual differences. For example, the concept "chair" is the knowledge of the essence "chair," which must be in every chair at all times and in all places, regardless of size, weight, color, material, and other individual differences.

The real object (a tree or a chair) and likewise the corresponding percept and phantasm, is individual, material, limited to a particular place and time; the concept is universal, immaterial, not limited to a particular place and time.

Only human beings have the power of intellectual abstraction; therefore, only human beings can form a general or universal concept. Irrational animals have the external and internal senses, which are sometimes keener than those of humans. But because they lack the rational powers (intellect, intellectual memory, and free will), they are incapable of progress or of culture. Despite their remarkable instinct, their productions, intricate though they may be, remain the same through the centuries, for example: beaver dams, bird nests, anthills, beehives.

One can find examples from philosophy that capture a deep awareness of this connection, to the point of fully understanding [the telos of Creation](#). The work of [Wynand de Beer](#) looks particularly promising in elucidating some of those connections, but there are many others. The "[Third Way](#)" and "[Third Evolutionary Synthesis](#)" of modern science are generally headed within the same direction.

Information creates a lattice in the mind for something to become manifest. I once tried to explain it to a friend like this:

I interpret the mind as using sensory data and feelings as the "substance" for the formulation of our mental conceptions (whether they be memories, dreams, etc.). This is part of the reason why I study so much. By taking in lots of information, it gives a lot of "substance" for intuition to play with and connections start to become apparent that would take a long time to uncover otherwise. The more subtle aspects of reality become "visible" to "the Mind's eye" when both the "idea" and the "material" are taken *together*.

The same is true collectively. We manifest circumstances, or entire civilizations, through the ideas that we entertain as a group. Kayla is right on it, and your description of the unicorn is brilliant: *“a sort of principality with a subtle body like an idea, that simultaneously coordinated learning in parallel over great distances”*.

What is being made through this coordination? Bob had mentioned how specific human thoughts were helping to shape a sort of “New Jerusalem” within the Void. Again, from page 43 in Chapter 4 - Clouds of *Fisherman’s Guide to the Cosmic Order*:

Suddenly there appeared in the void a city of light in brilliant color. It was suspended in the void in such a way that the underside of the city along its nearest edge was at first visible, and there were dark tendrils of energy reaching down into the void, like roots from which it drew its sustenance. The city was recreated from the energies of the void, its variety selected and assimilated at will by the Being behind me.

Then my perspective rose to provide a clear view across the city and down into the streets on the nearest side below me. It was a beautiful city, immaculately clean, with cobblestone streets and mostly masonry-type buildings in a variety of pastel shades, none of which were over a few stories high. There were both peaked and flat roofs, an old English style house on a corner, domed structures, and varied shaped buildings extending over quite an area, but it was not modern or as large as many cities are today. It tended to have a Mediterranean character, sort of like a new Jerusalem, although it had no specific resemblance to Jerusalem, and no churches, temples or mosques were noticed. It was brilliantly illuminated in a rich mosaic of colors, with no people or vehicles apparent. The city appeared to be empty.

Awhile back, I had mentioned that I wanted to form online communities for collaborative creative problem-solving of world issues, and for the building of ecological makerspaces to implement the solutions that are generated through them...They would probably be something like “unicorn forests”. ☺

This is one aspect of a larger plan. Many aspects of that plan are proceeding in parallel, but in different forms and at different rates, so that if any one fails, hopefully they are diverse enough to persist through time.

On the software-side, I am trying to develop a tool that I call [RNDTBL](#) (although much of the planning behind it is missing from that page). On the hardware-side, I am trying to find ways to simplify [the production of computers](#) to put them within the reach of home experimenters. This includes not only chip fabrication, but also the tools for networking.

An adjacent goal was to compile databases with the [DIY information necessary to rebuild society from scratch](#) in ways that truly are sustainable. Then, this would be shared through a worldwide mesh network formed of [packet radio](#) transmitters

and receivers...It was intended to be something like a “second Internet” that is not dependent upon the backbone, something that would be fault-tolerant, [disaster-proof](#), and work with very little underlying infrastructure. No cables, all air. It would be comprised of homemade devices that can be created and powered in a multitude of different ways. This would give everyone access to it.

The bottleneck is the limitation of data transfer speeds in packet radio. However, one can [twist light and radio waves to be able to hold more data](#) (e.g.: experiments have accomplished up to 2Tb/sec. with light and 32Gb/sec. with radio waves). Therefore, is it possible to create a relatively cheap vortex-shaped antenna or waveguide that would make every packet radio user into their very own ISP? I feel that the incredible breakthroughs that arose from amateur radio and home computing are far from over, especially when we get back to their DIY roots and [conviviality](#).

But despite the wonder that I feel about our ultimate home, I am also extremely cautious about how we get there. Our “tool systems” have to amplify the constructive within our “human systems” in order for the transition to be as smooth as possible. As it now stands, [they are not doing that](#). Building a bridge to heaven is less about the technology that we wield, and more about the compassion that we express towards All things.

Through Agape, [The Logos](#) is made manifest.  
[The Infinite Light](#) appears within the Heart of a human being.

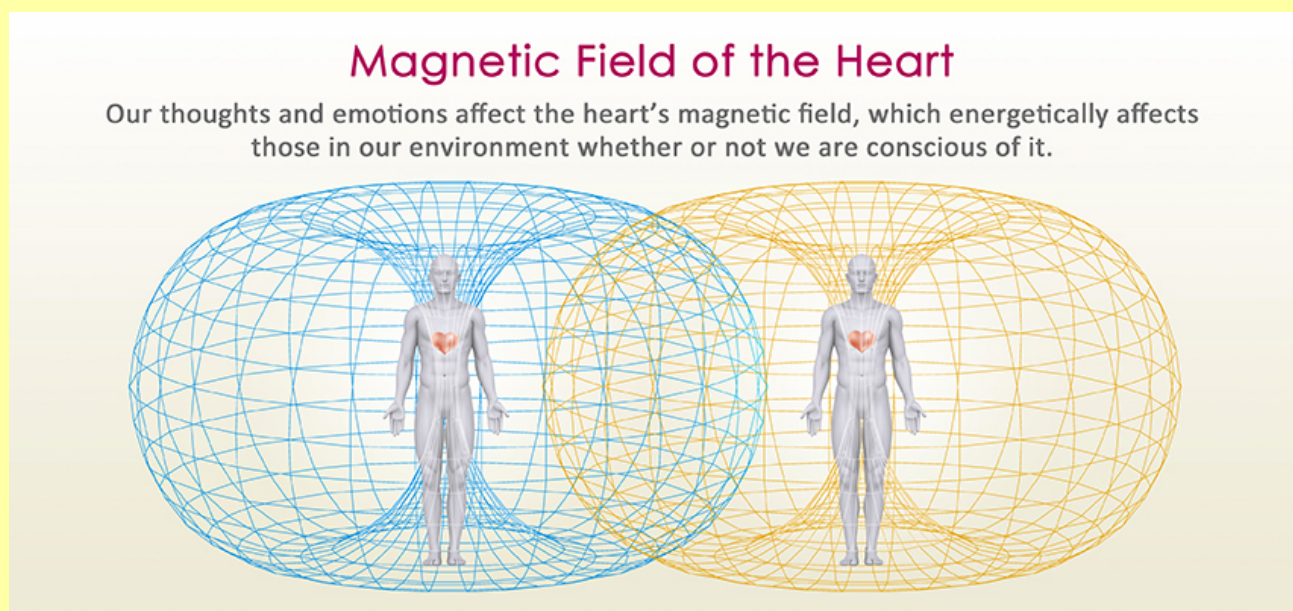


image from [The HeartMath Institute](#)

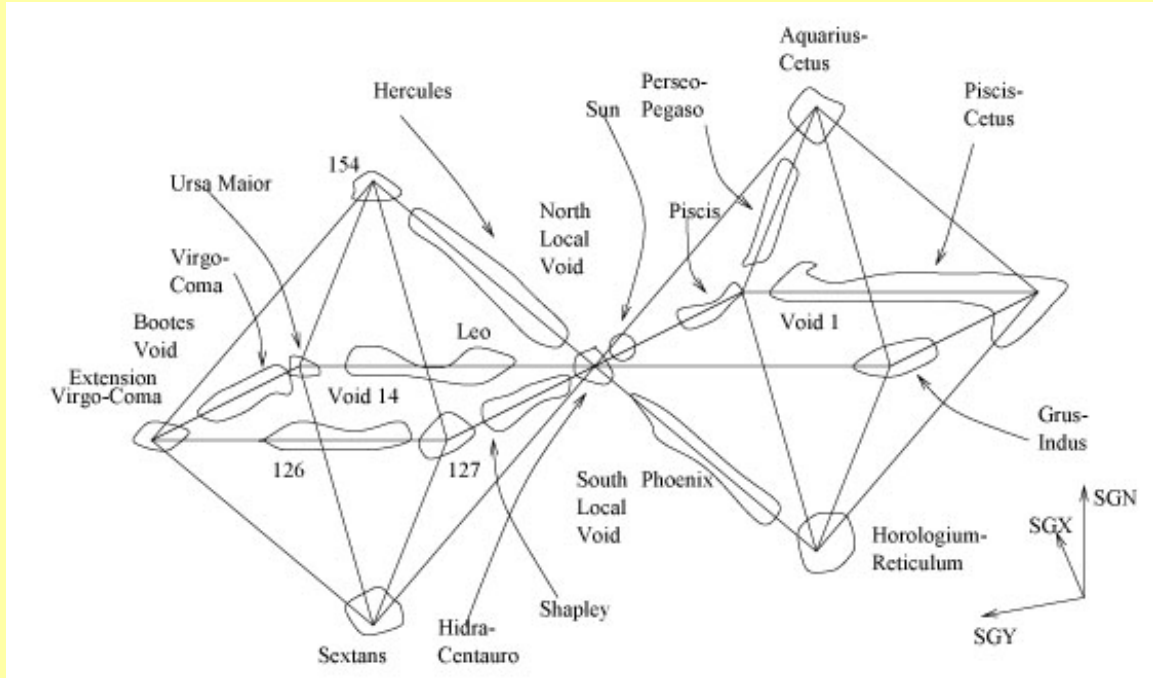
Upon your first mention of the unicorn, I was struck by how you said that you were: *“trying to map at least 5 complementary perspectives onto the helical rings that cycle through the 4D polytope of the 120-Cell”*. Can you elaborate on this?

Immediately, I had thought of some research that I had heard about long ago, [The Poincaré Dodecahedral Space Model Gains Support to Explain the Shape of Space](#) by Jean-Pierre Luminet, Jeffrey R. Weeks, et al.

I had first come across this in an article from a (now sadly defunct) website, [MIQEL](#). I will quote it here [with some slight editing and extra links by me]:

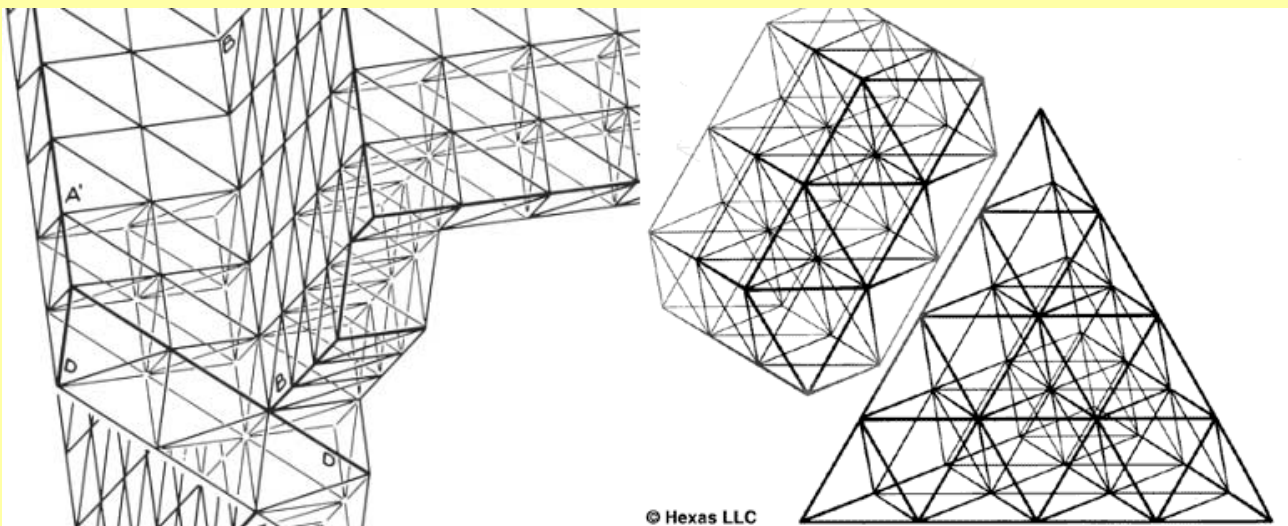
**Amazingly, [recent galaxy-cluster mapping](#) has shown geometry!  
The local voids outline Close-packed Octahedra!!!**

This seems stunning, but is probably due to gravitational / expansive forces finding equilibrium in minimum-energy vectors. If we have 2 octahedrons (and presumably more in a 3-D grid) then we also have tetrahedral voids, because octahedrons don't fill-space (like stacking cubes). It requires alternating *Octa-Tetra* combination to fill space in this type matrix. More on the surprising geometry of this pattern below ...



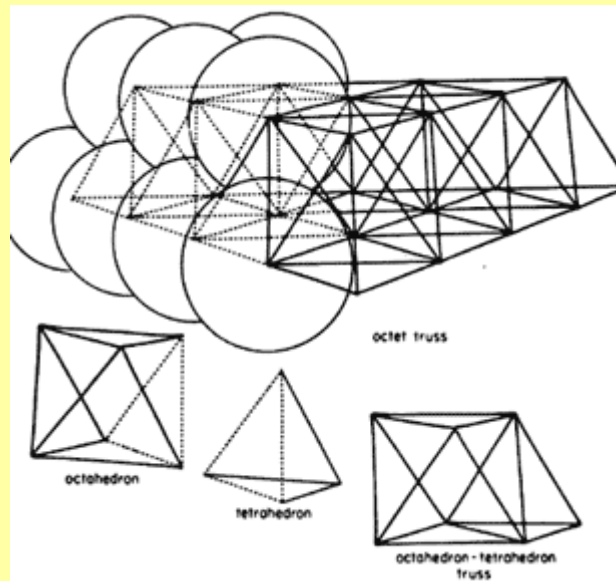
**Below:  
This is the full pattern implied by the supercluster geometry,  
it's what is called a '[Space Frame](#)' trusswork.**

It was discovered by visionary genius Buckminster Fuller and patented in 1961. One amazing feature is that in examining the matrix **you can find all the other platonic solids outlined except those with 5-symmetry**. Look at these different angles of the oct-tet matrix and notice you can see outlines of octahedra, tetrahedra, truncated tetrahedra, nested tetrahedra, cubes, cubes with tetrahedra inside, rhomboids, and more ...



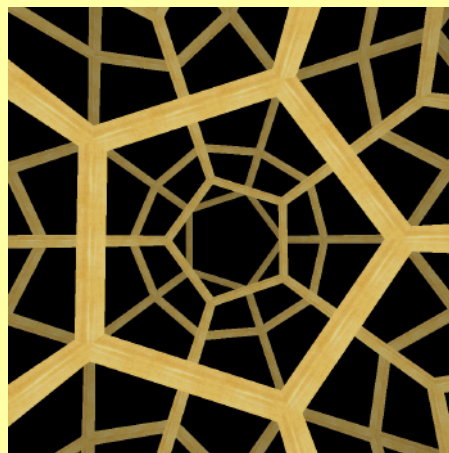
It was later discovered that Alexander G. Bell has come upon the same structure in 1903, but Bell was looking to build a practical lightweight platform & didn't notice the deeply fundamental nature of the spaceframe as the maximum self-reinforcing structure possible in platonic 3-D geometry.

**The most amazing thing is this arrangement happens anytime spheres are packed together.** It's the most economical, space/energy conserving arrangement of equal size spheres. Spheres pack 12 around 1 - each sphere in the matrix is touching 12 other spheres and connecting grid-lines (force vectors) from the center of each sphere creates our space-frame trusswork.



Microwave data gathered by the WMAP satellite indicates the **Universe may be in the shape of a 'chiral' Dodecahedron - a twelve sided polyhedron with pentagonal symmetry.**

So ... My first question is how does the **6-symmetry space-frame superclusters** matrix relate to the **5-symmetry dodecahedral universe** 'shape' (not proven ... still hotly debated)!  
hmmm ... the spheres pack 12-around-1 and the Dodeca has 12 faces.  
Just an observation ...



More about the possibly **dodecahedral universe** here -  
[http://news.nationalgeographic.com/news/2003/10/1008\\_031008\\_finiteuniverse\\_2.html](http://news.nationalgeographic.com/news/2003/10/1008_031008_finiteuniverse_2.html)

The “12-around-1” archetype is important. To reiterate, by taking a cluster of 13 close-packed spheres and connecting their centers, we get a cuboctahedron:

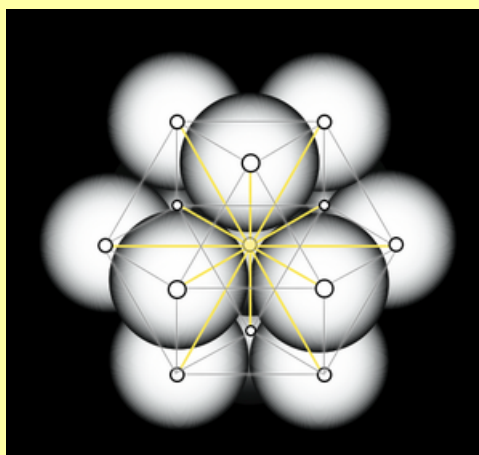


image from [Cosmometry](http://www.cosmometry.net)

Buckminster Fuller did not interpret geometric forms as immutable “solids”. Instead, he treated them as dynamic networks of forces, so every edge is a vector. He refers to the cuboctahedron as a “Vector Equilibrium” because he saw the 12 vectors raying out of the center of the central sphere (shown in yellow within the above image) as being constrained by those around them, holding all of the forces that it describes in equilibrium.

He would also make models of the cuboctahedron out of sticks connected by flexible rubber connectors so that the whole structure was compressible. This led to his concept of the “[Jitterbug](http://www.cosmometry.net)”, where the 24 edges of the cuboctahedron twist and collapse to form other geometries as it is compressed.

An interesting property of the cuboctahedron is that its edges can be thought of as four hexagons 60-degrees away from one another in space:

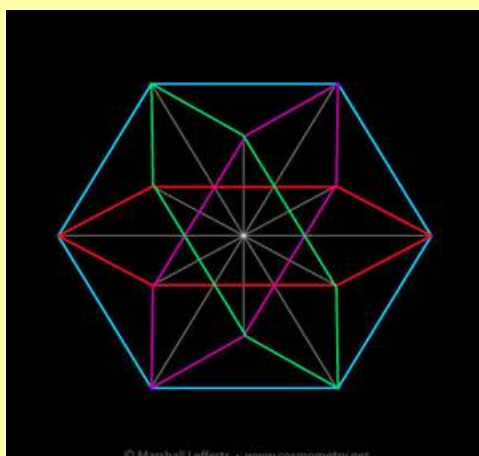
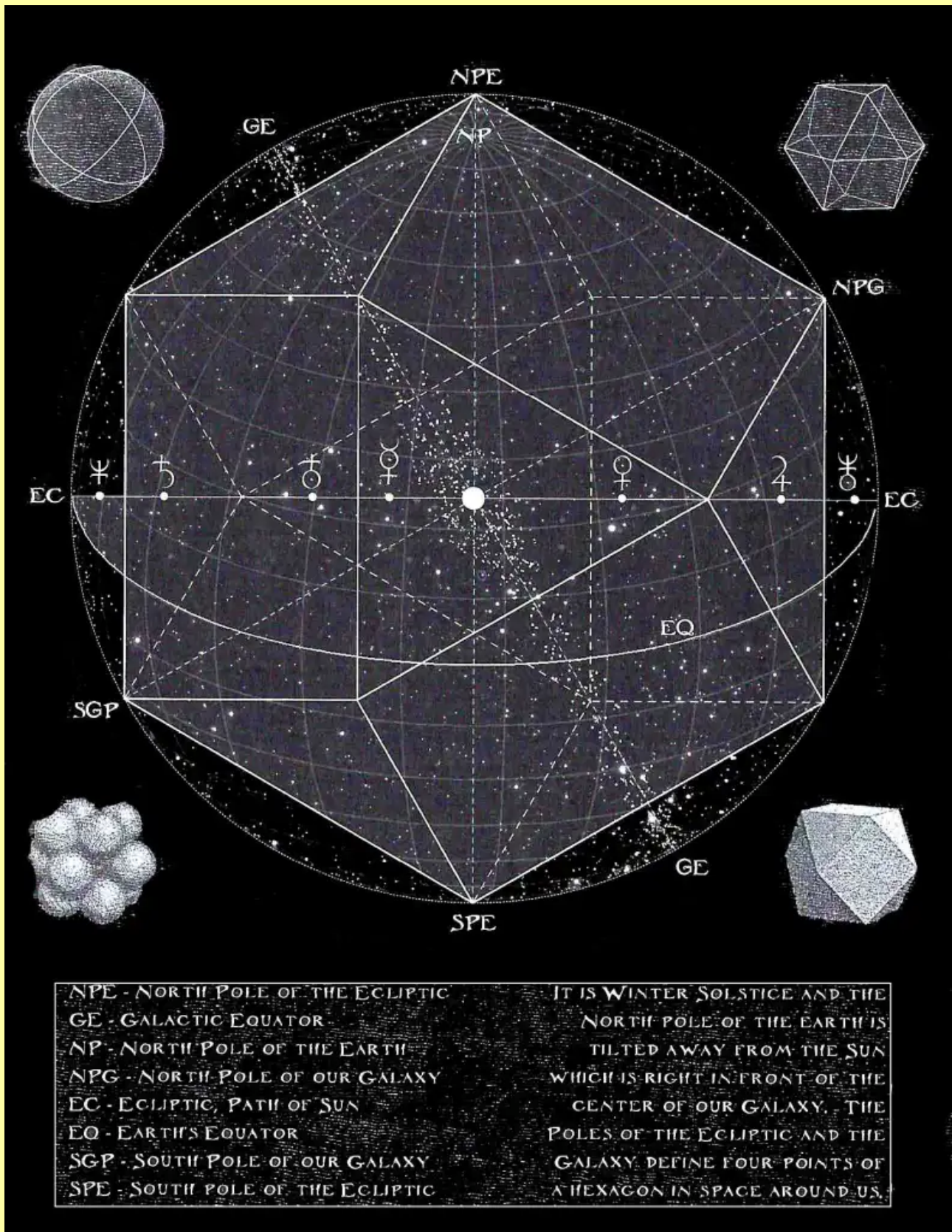


image from [Cosmometry](http://www.cosmometry.net)

This is “Nature’s Coordinate System”, four axes at 60-degrees, rather than three axes at 90-degrees like the traditional Cartesian coordinates. Indeed, there seems to be an underlying cosmic logic to it. To paraphrase *A Little Book of Coincidence* by John Martineau [also on pg. 356 of *Quadrivium* by Wooden Books]:

“One of the most obvious symmetries of modern cosmology occurs in that the Milky Way, i.e. the plane of our own galaxy, is tilted at almost exactly  $60^\circ$  to the ecliptic or plane of our solar system (the horizontal line in the image below) [99.7 %]. What is more, every year the Sun crosses the galaxy through the galactic center, and being alive in these times means this happens on midwinter’s day.”



If we turn these four hexagons into great circles on the surrounding sphere, then we end up with a spherical cuboctahedron. A little known plant geneticist by the name of Derald Langham refers to this figure as a “Genesa Crystal”:

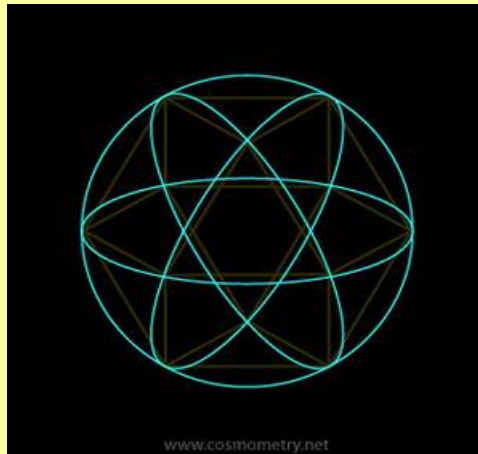


image from [Cosmometry](http://www.cosmometry.net)

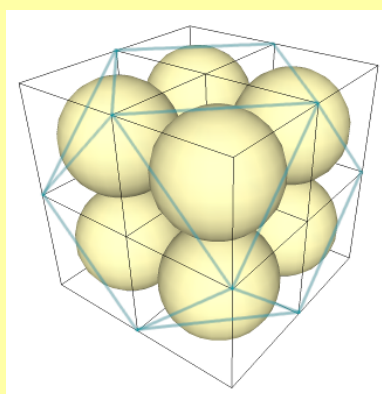
He derived it from his observations of embryos and the fact that the first three cell divisions of every multicellular creature seem to be the same:



image from [Cool Science](http://www.coolscience.org)

It is the familiar doubling of dimensions: One cell (point) divides into two. Two cells (line) divides into four. Four cells (square) divides into eight. Three divisions.

The first eight cells (cube) create a structure that he calls “4-over-4” from which the cuboctahedron can be extracted:



I tried to give an idea of the presence of the cuboctahedron in blue.

Since the cells only start to [differentiate](#) after that point, he thought that this figure was symbolic of infinite potential. Notice that we've used this geometry on two vastly different levels of scale (i.e.: embryonic and solar-galactic).

Please keep in mind that, in the same way that we spoke of The Planet Mind, [the Solar System is also an organism](#). And so on up "[The Great Chain of Being](#)", until we get [all the way back to The Supreme Intelligence](#), in whose image we are made.

We exist inside of The Eternally Self-Existent One,  
always have and always will.

Derald Langham was an interesting character. He would make giant spherical cuboctahedrons, take them out into the forest, and then meditate within them:

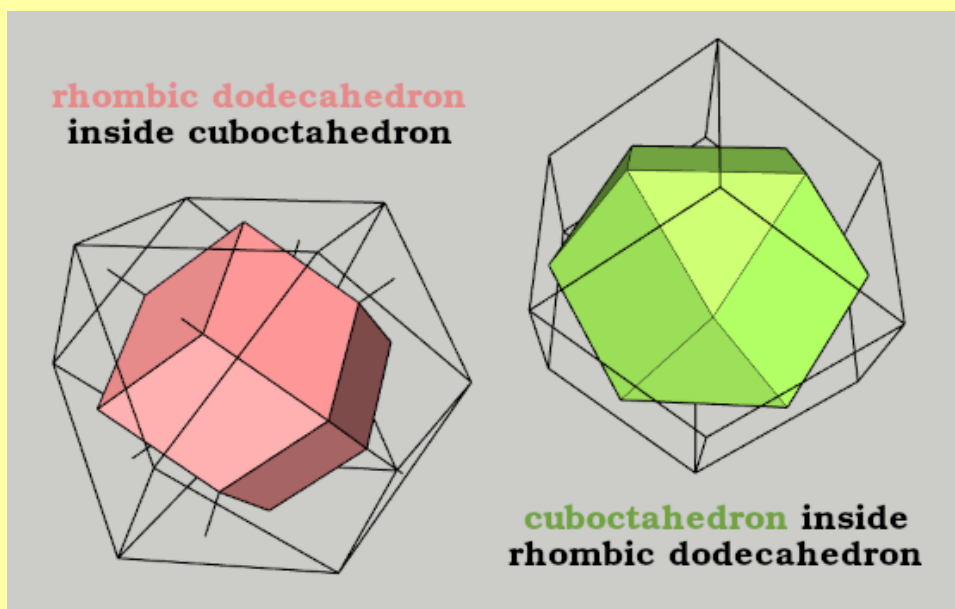


image from [Genesa](#)

From this, he was inspired to write books with seemingly enigmatic titles, such as [Genesa: A Conceptual Model to Synthesize, Synchronize, and Vitalize Man's Interpretation of Universal Phenomena](#) and *13-Dimensional Genetics: Tomorrow's Thinking Today*. I encourage you to make [a small paper model](#) and tell me if you notice anything different about the interior space in contrast to the surrounding environment.

Now, let's see if we can extend the geometry...

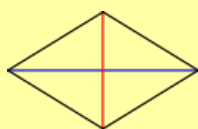
If we take the dual of the cuboctahedron (i.e.: turn its faces to vertices, or vice versa), then we get a [space-filling polyhedron](#) known as a rhombic dodecahedron:



VBM considers the fundamental structure of the Universe as made up of rhombic dodecahedra, and there is at least [one other model](#) that I know of that approaches it in a similar way. The same concepts can be connected to other models through [various geometric transformations](#).

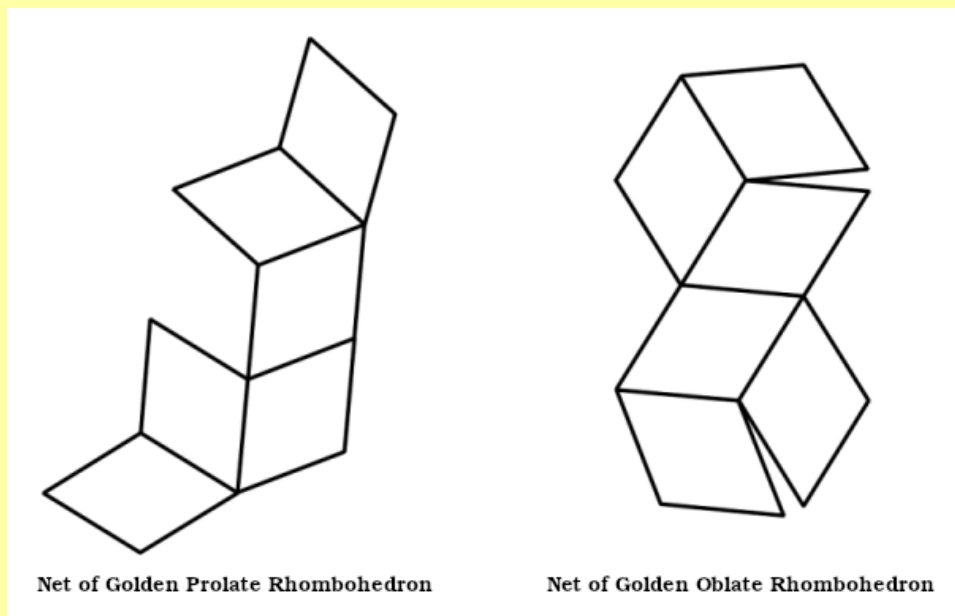
It seems likely that The Golden Mean Proportion was sacred to the Pythagoreans because it was evidence of [the psyche of the Cosmos](#), a transcendent pattern that united All things in Harmonia.

Let's turn each of the rhombic faces of the rhombic dodecahedron into a golden rhombus whose diagonals are in a Golden Mean Proportion:



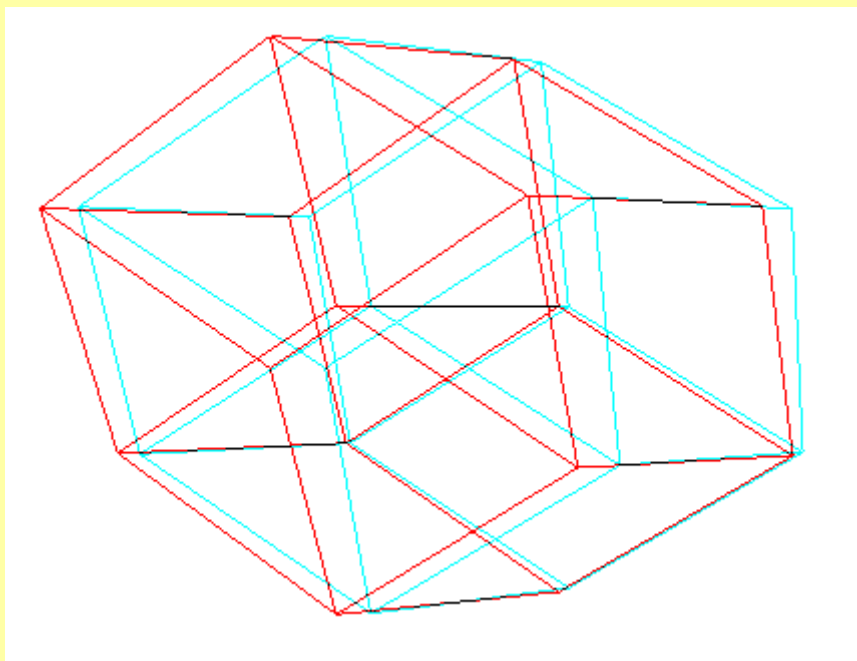
Self-similarity along twelve axes simultaneously! The polyhedron that we end up with is sometimes referred to as a “rhombic dodecahedron of the second kind” or a “Bilinski dodecahedron”. It is part of a whole set of golden polyhedra, each of which can be derived from [two of them](#):

- \* A skinny one referred to as a golden prolate or acute rhombohedron
- \* A flat one referred to as a golden oblate or obtuse rhombohedron



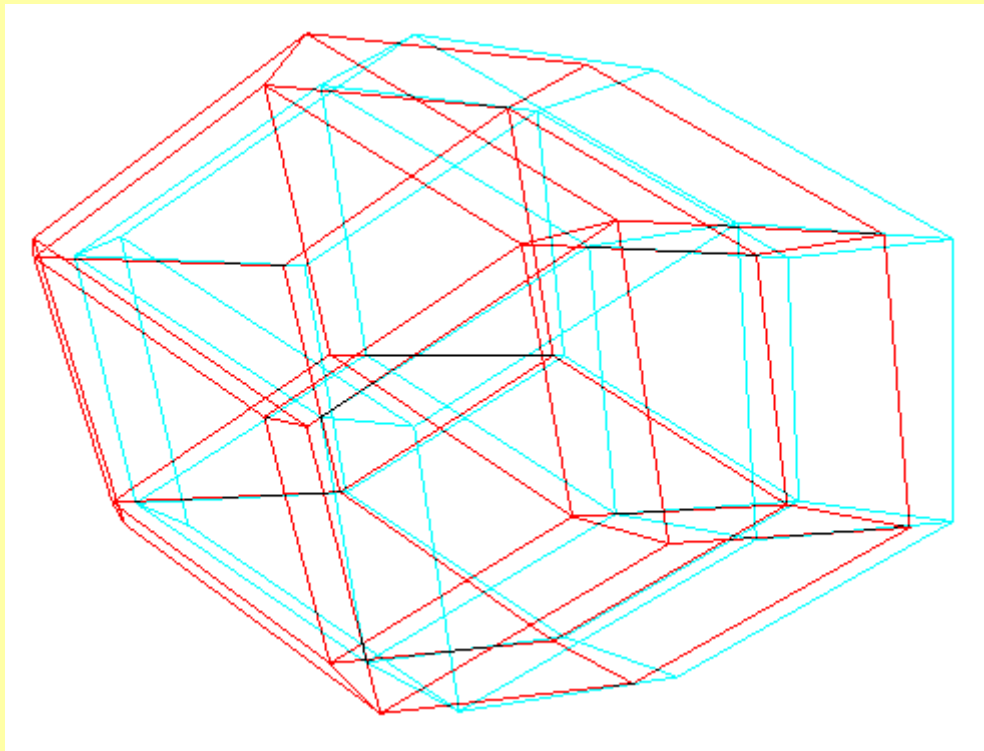
These fill space like a three-dimensional [Penrose tiling](#), and can be combined in various ways to give us the different “[isozonohedra](#)”. To demonstrate...

\* two golden prolate rhombohedra + two golden oblate rhombohedra = one rhombic dodecahedron of the second kind (12 faces)



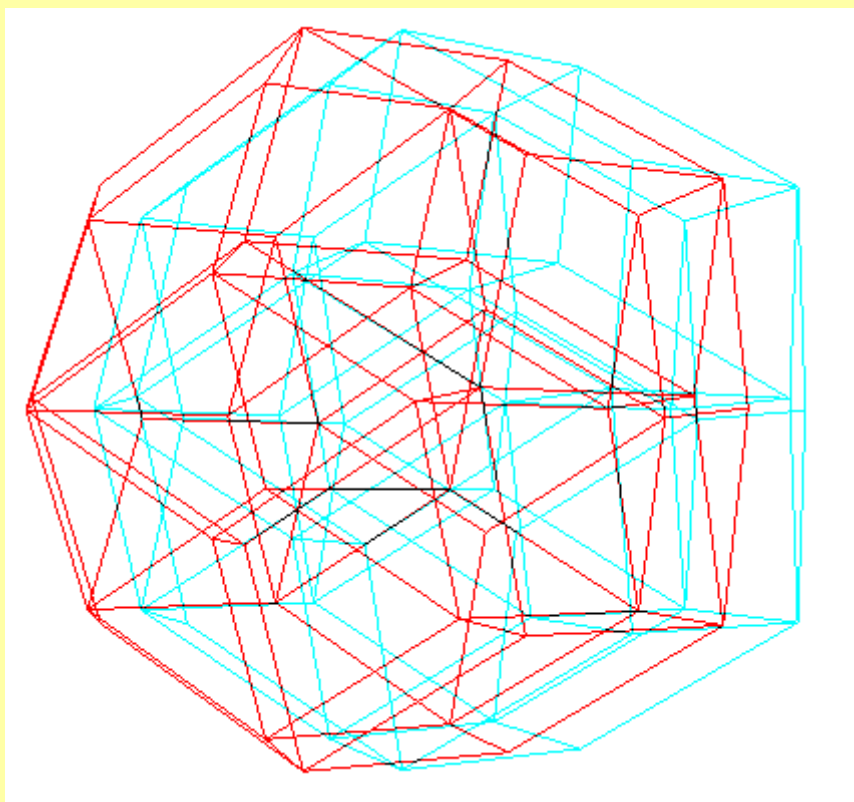
anaglyph from [Robert Inventor](#)

\* five golden prolate rhombohedra + five golden oblate rhombohedra = one golden rhombic icosahedron (20 faces)



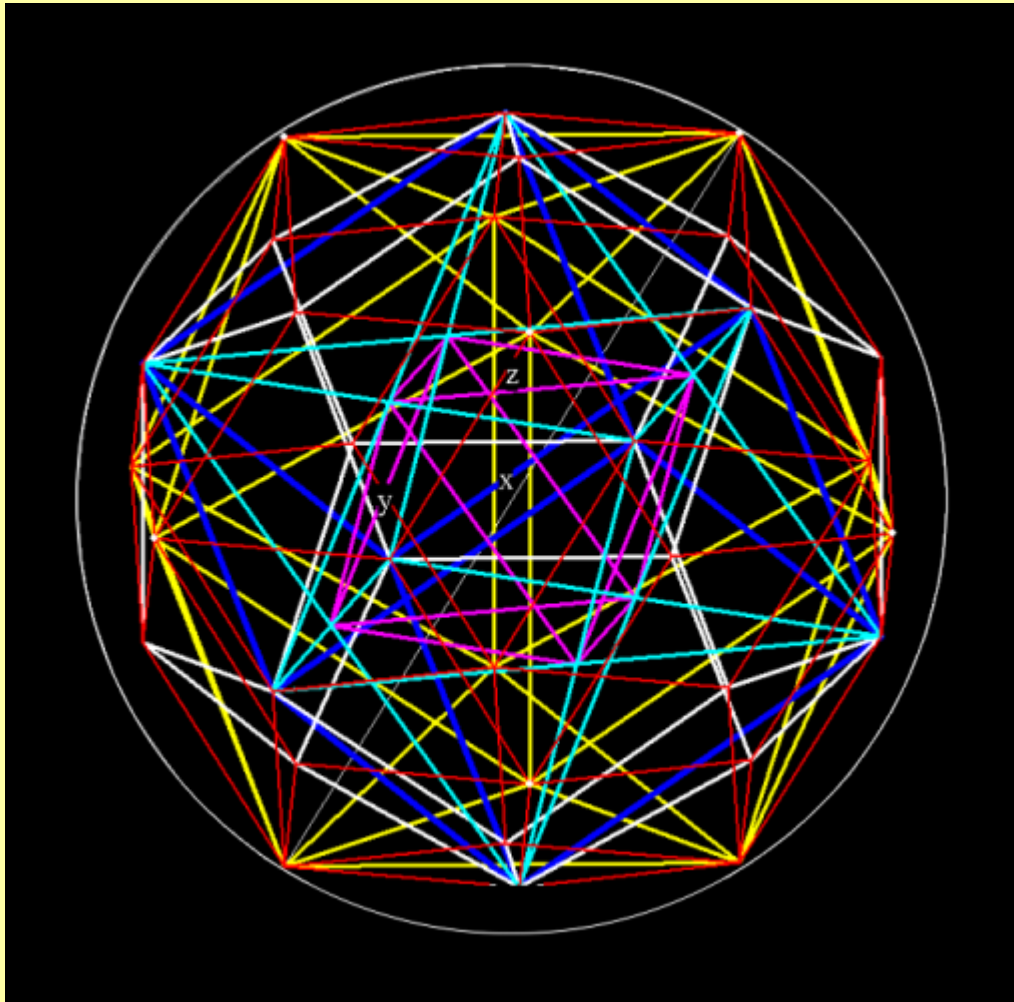
anaglyph from [Robert Inventor](#)

\* ten golden prolate rhombohedra + ten golden oblate rhombohedra = one rhombic triacontahedron (30 faces)



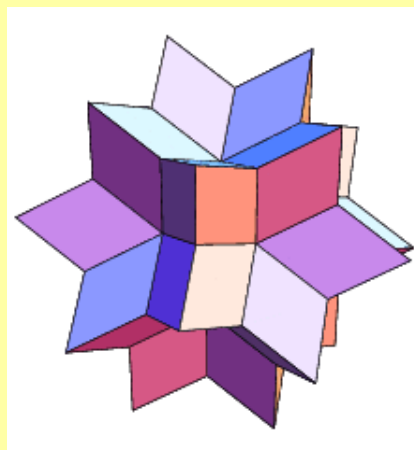
anaglyph from [Robert Inventor](#)

Similarly, we can recursively nest the Platonic solids so that their edge lengths repeatedly make Golden Mean Proportions. [Part of this pattern](#) shows up within a sacred geometry figure known as “[The Lesser and Greater Mazes](#)”, but all of the Platonic solids are nested in this way inside of the rhombic triacontahedron!



This image (from [K.J. MacLean](#)) shows all five Platonic solids nested within the rhombic triacontahedron simultaneously. An octahedron is in magenta, a star tetrahedron in cyan, a cube in blue, a dodecahedron in cream, an icosahedron in yellow, a rhombic triacontahedron in red, and an outer sphere in white.

Finally, if we take twenty golden prolate rhombohedra, we can combine them to make a rhombic hexecontahedron (60 faces):



We can think of the interior of the rhombic hexecontahedron as containing a rhombic triacontahedron:

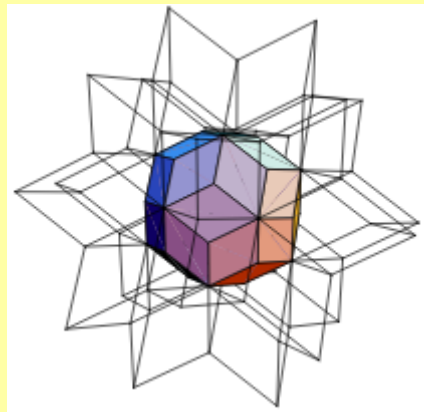


image from [Wolfram MathWorld](http://mathworld.wolfram.com/RhombicHexacontahedron.html)

The concave surface of the rhombic hexecontahedron matches the convex surface of the rhombic triacontahedron, so they fit like hand in glove...

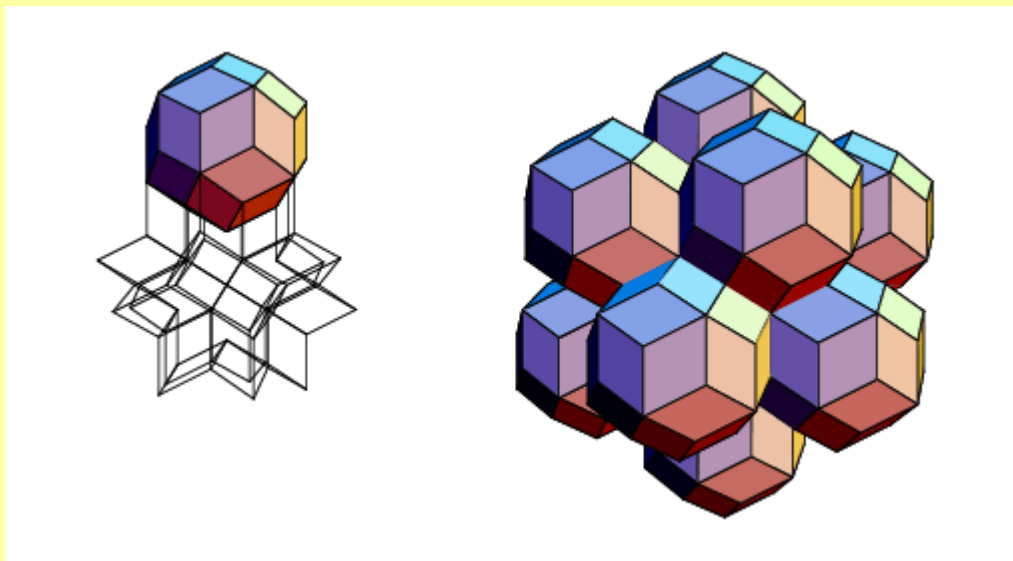


image from [Izidor Hafner](http://www.izidorhafner.com/)

...and they pack 12 around 1!

To quote Exodus 33:19-20:

And יהוה said, "I will cause all my goodness to pass in front of you, and I will proclaim my name, יהוה, in your presence. I will have mercy on whom I will have mercy, and I will have compassion on whom I will have compassion. But," he said, "you cannot see my face, for no one may see me and live."

In 1987, Lynnclaire Dennis died from a lack of oxygen due to a hot-air balloon ride that went up too high. At the apex of her near-death experience she, [quote](#):

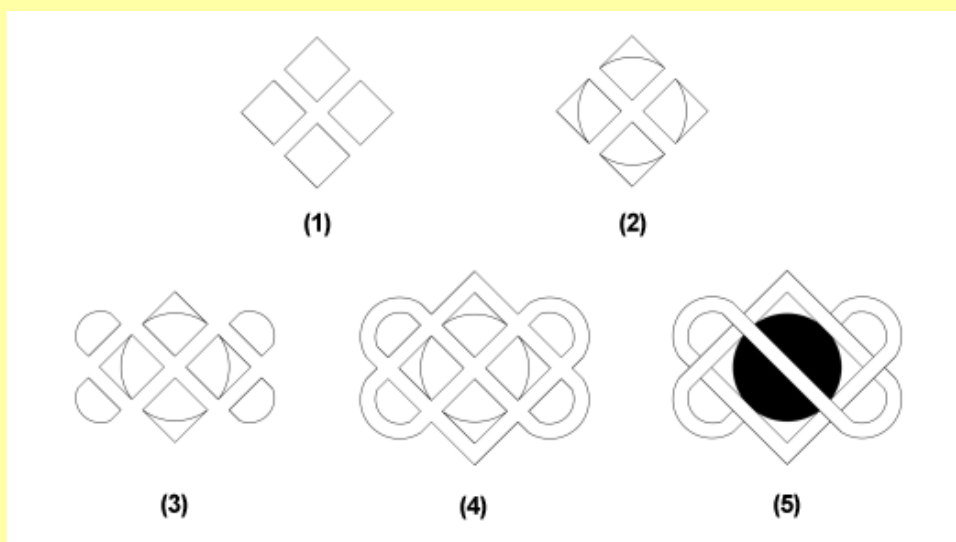
I arrived at the pinnacle and, standing at the entrance to the Light, took a single step, leaving my right footprint embedded in Eternity. I entered a sacred space - a place where I knew I had returned to my most essential nature, where I felt wholly and consciously united with all things and Source, where a soothing balm of peace was poured on my spirit by an unseen hand, an emollient so rich in love that to this day I cannot fully absorb or comprehend it.

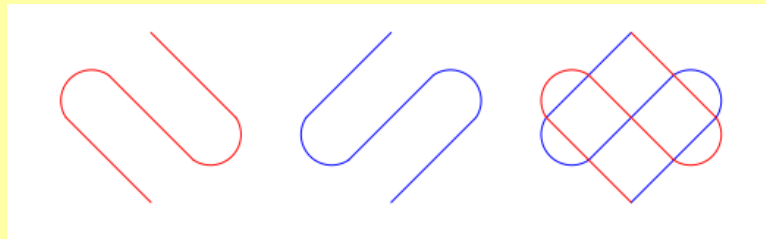
And then, in one ephemeral glimpse, I saw the Pattern, the single strand of the tapestry I knew was the essence woven through matter in every reality. Its design was so complexly simple that I knew it could only have been fashioned in the exalted intricacy of infinity.

Seeing the Pattern, I knew I was looking at life itself. It was light; it was time and space. It was the energy of all matter, the heart of all that mattered. It was the very essence of all being. It emanated from Source, illuminated to my mind by "the Source behind the sun" as it moved in perfect harmony with all the universe. As I prepared to meld into the Source of Light and absolute Love, I knew with all my being that the Pattern was the core of all substance. I knew that the MUSIC emanating from the Pattern was the song of my heart, a testament of unconditional Love. The single step I had taken was the first in a dance that would take me into the single point of Infinite Light, which contained the power of Love that would forever illuminate my mind and heart...

Interestingly, after repeated experiences, she eventually brought back [an understanding of the "Pattern" that is mathematical in nature.](#)

To give [a very simplified description,](#) it is [a specific nesting of polyhedra](#) surrounded by [a type of trefoil knot.](#) The polyhedron is related to the rhombic triacontahedron, whereas the knot is related to a torus.





### Six Perspectives of The Pattern\*



THE HEART  
OF INTIMACY



GODDESSENCE



THE NUCLEUS



THE RAINBOW  
BRIDGE



THE BLOSSOMING  
OF PEACE

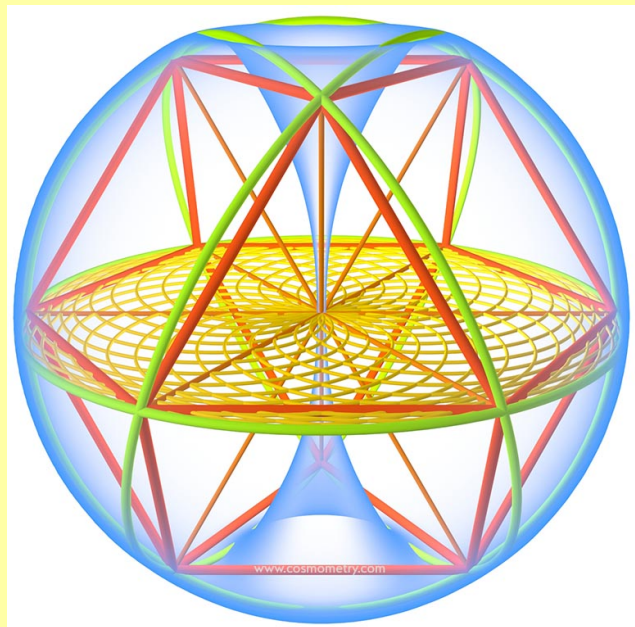
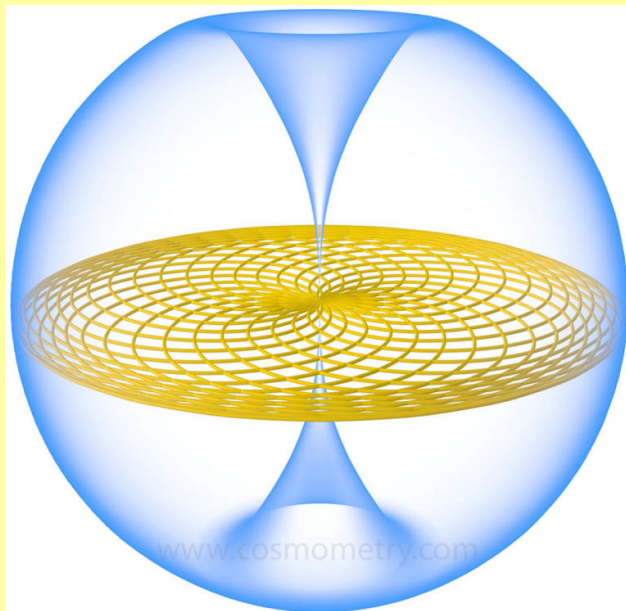


THE FIELD  
OF FORGIVENESS

\*Lynclaire named these six initial perspectives of the Pattern as she became aware of each one. These and other perspectives of the Pattern have seemed to correlate with levels of awareness she has experienced as part of her inner growth.

The above images are adapted from the book *The Pattern* (pgs. 99, 122, and inner cover)

A helpful visualization of how the two interact is to imagine the cuboctahedron inside of a torus, something like this:



images from [Cosmometry](http://www.cosmometry.com)

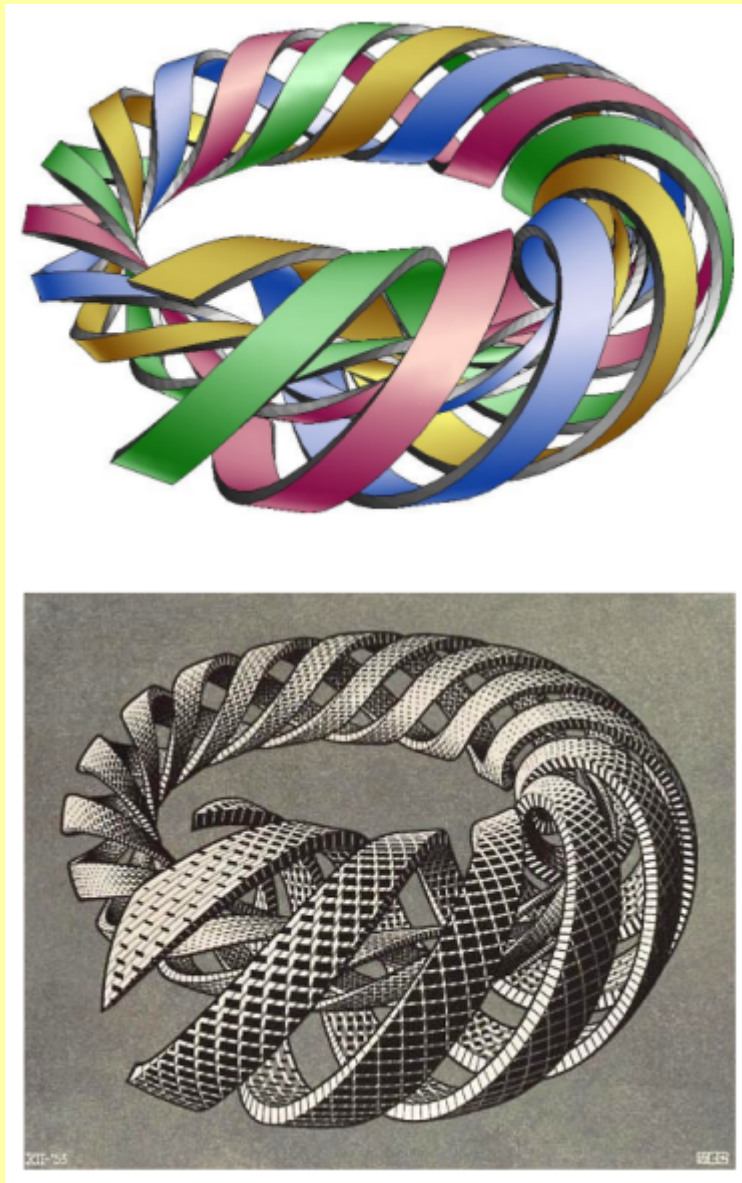
We've been guided to a path back to the very heart of The One.

## Addenda:

Thank you for *The Initiates of the Flame* booklet. Manly P. Hall is always a delight to [listen](#) to or read.



The alchemical symbolism of the ouroboros has always stood out to me. I was given the interpretation of the snake eating itself as a symbol of a self-sustaining cycle, of nourishment being channeled towards ever increasing levels of nourishment. And it was represented mathematically as a torus of constantly increasing diameters, like M.C. Escher's "Spirals":



Things can be eternally regenerated, on the individual level...

...It was a recognition that possibilities need not be confined in any way, that the range of possibility is unlimited. I seemed able to see forever into the depths of the shining void. This was followed by a few examples, most of which I cannot find words to describe. One of them, a little frightening in its implications, was the possibility of unlimited lifespan. This brought with it a trace of anxiety about how to relate to an indefinite lifespan when our thoughts are so conditioned to a brief lifetime of striving ending in death... (from page 35 of Chapter 4 - Clouds in *Fisherman's Guide to the Cosmic Order*):

...and on the collective level...

...Are you saying then that we can function indefinitely, with complete security and stability, without accumulating any profit whatever?...(from page 66, Chapter 8 - Profit Distribution for Survival Advantage in *Enlightened Management*)

God willing, the next document will try to apply Systems 4 and 5 to the human body and the biosphere to do exactly that.

The above geometric explorations are related. The fundamental levels of space and time can be reduced to [patterns in whole numbers](#). Likewise, there is [a whole parallel history of chemistry](#) that is connected to its alchemical roots. Biological transmutations are ubiquitous (e.g.: [within the mitochondria](#)). Unfortunately, the Involutionary Variant keeps driving people towards applying this sacred science towards self-destruction (e.g.: [the production of atomic weapons](#)).

However, I remain steadfast in my hopefulness. Love transforms everything.  
I hope you are doing well!

-Sage